DR. BABASAHEB AMBEDKAR TECHNOLOGICAL UNIVERSITY, LONERE

Regular & Supplementary Summer - 2024

Course: B.Tech. **Branch :** Computer and Allied **Semester : IV**

Subject Code & Name: BTBSC404 Probability and Statistics

Max Marks: 60 **Date: 20/06/2024** **Duration: 3.00 Hr.**

Instructions to the Students:

1. All the questions are compulsory.
2. The level of question/expected answer as per OBE or the Course Outcome (CO) on which the question is based is mentioned in ( ) in front of the question.
3. Use of non-programmable scientific calculators is allowed.
4. Assume suitable data wherever necessary and mention it clearly.

(Level/CO) Marks

Q. 1 Solve Any Two of the following. **C0 1**

A) State and prove Multiplication theorem of Probability. **6**

B) An urn contains 8 white and 3 red balls. If two balls are drawn at random, find the probability that 6
(i) Both are white (ii) both are red (ii) one is of each colour

C) In a bolt factory, machines A,B,C manufacture respectively 25%, 35% and 40% of the total. Of their total output 5, 4, 2 per cent are known to be defective bolts. A bolt is drawn at random from the product and is found to be defective. What are the probabilities that it was manufacture by **6**
(I) Machine A (II) Machine B or C

Q.2 Solve Any Two of the following. **CO 2**

A) Find the Mean, Variance and Standard Deviation of the discrete random variable x whose probability distribution is given as **6**

X	1	2	3	4	5
P(x)	0.1	0.1	0.3	0.3	0.2

B) A car hire firm has two cars which it hires out day by day. The number of demands for a car on each day is distributed as a Poisson variate with mean 1.5. Calculate the proportion of the days on which **6**
(i) Neither Car is used (ii) Some Demand is refused.

C) The hourly wages of 1,000 workmen are normally distributed around a mean of Rs.70 and with a standard deviation of Rs.5. Estimate the number of workers whose hourly wages will be: **6**
i) Between Rs. 69 and Rs. 72 ii) more than Rs. 75 iii) Less than Rs. 63
[(Area between Z=0 and Z=0.4)=0.1554, (Area between Z=0 and Z=0.2)=0.0793, (Area between Z=0 and Z=1)=0.3413
(Area between Z=0 and Z=1.4)=0.4192]

Q. 3 Solve Any Two of the following. **CO 3**

A) From the following data calculate Karl Pearson's correlation coefficient. **6**

x	6	2	10	4	8
y	9	11	5	8	7

Also find standard error (S.E) and probable error (P.E).

B) From the following data calculate rank correlation coefficient. **6**

x	32	35	39	60	43	37	43	49	10	20
y	40	30	70	20	30	50	72	60	45	25

C) Prove that Correlation Coefficient r is independent of change of origin and scale. **6**

Q.4 Solve Any Two of the following. **CO 4**

A) Obtain the equation of line of Regression of y on x and x on y for the following data. **6**

x	1	2	3	4	5	6	7	8	9
y	9	8	10	12	11	13	14	16	15

Also find the value of y when x = 6.2

B) The lines of regression of a bivariate population are 8x-10y+66=0 and 40x-18y=214. The variance of x is 9. Find (i) Mean values of x and y (ii) Correlation coefficient between x and y (iii) Standard deviation of x and y. **6**

C) At the time of estimation of the regression equations of the two variables x and y ,the following results were obtained: $\overline{x} = 90$; $\overline{y} = 70$; $n = 10$; $\sum x^2 = 6360$; $\sum y^2 = 2860$, $\sum xy = 3900$, where x and y are the deviations from the respective means. Obtain the equations. **6**

Q. 5 Solve Any Two of the following. **CO 5**

A) In a random sample of 400 persons from a large population, 120 are females. Can it be said that males and females are in the ratio 5:3 in the population? Use 1% level of significance. **6**

B) It is claimed that a random sample of 100 tyres with a mean life of 15269 kms is drawn from a population of tyres which has a mean life of 15200 kms and a standard deviation of 1248 kms. Test validity of claim at 5% level of significance. **6**

C) In two large populations there are 30% and 25% respectively of fair haired peoples. Is this difference likely to be hidden in sample of 1200 and 900 respectively from the two populations? Use 5% level of significance. **6**

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